A Simulation Based Approach for Exploring the Change of Support Problem and Related Questions

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1. Areal data and change of support

- Areal, or region-indexed, spatial data are common in environmental, ecological, and public-health settings.
- Spatial dependence may be represented using distance-based covariance functions or neighborhood-based models such as CAR and ICAR Gaussian Markov random fields.

The **change of support problem** arises in settings where spatial data

- are observed at one resolution but inference concerns another,
- combine point-referenced and areal observations, possibly with areal data at multiple resolutions.

The **centroid approximation** is a simple workaround: treat each areal value as a point observation at its polygon centroid. This ignores averaging over its spatial support; for irregularly shaped units, centroid-to-centroid distance may also poorly represent spatial proximity. For example, it can obscure the long shared boundary between Chile and Argentina.

Applied question: does centroid-based inference about spatial structure depend on the grid resolution of an Earth-science data product?

For settings in which block data can sensibly be viewed as averages of an underlying point-referenced process, Gelfand et al. (2001) instead formalize

$$Y(A_i) = \frac{1}{|A_i|}\int_{A_i} Y(s)\,ds, \tag{1}$$

where $Y(s)$ is a latent continuous spatial process and A_i is the support of the ith observed areal value.

To approximate the integral on a fine grid:

- Let $\mathbf{y}_L$ contain process values at latent locations $\mathcal{S}_L = \{s_1, \ldots, s_N\}$.
- Let K be an $n \times N$ matrix whose ith row averages the latent locations contained in A_i.

$$\mathbf{y}_A \approx K\mathbf{y}_L. \tag{2}$$

In Simulations 1 and 3, data are generated directly as $\mathbf{y}_A = K\mathbf{y}_L$, so this equality is exact within the discrete simulation.

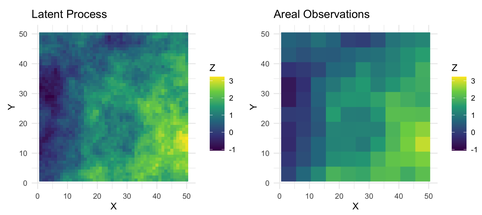


Figure 1. A latent realization and the observed block averages it induces.

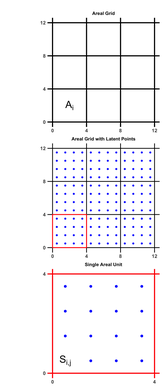


Figure 2. Observed unit A_i, latent locations s_{ij}, and two grid resolutions.

2. The covariance model used for estimation

- Assume the latent-grid process has Matérn covariance $\mathrm{Cov}(\mathbf{y}_L) = \Sigma_\theta$, where $\theta = (\sigma^2, \rho, \nu)$.
- Within the fine-grid approximation, aggregation gives

$$V_\theta = \mathrm{Cov}(\mathbf{y}_A) = K\Sigma_\theta K^\mathsf{T}. \tag{3}$$

- **All four simulations are fit using this same covariance model**, $V_\theta = K\Sigma_\theta K^\mathsf{T}$.
- Only Simulation 1 is generated from this model with a Matérn Σ_θ. Simulations 2–4 have different generating covariances for the observed data.

3. What the four simulations isolate

These simulations cross two generating covariance families with two meanings of "changing resolution." Every dataset is fit with the same areal model, $\mathbf{y}_A \sim N(\mathbf{0}, K\Sigma_\theta K^\mathsf{T})$; they do not yet compare it with a centroid-based likelihood.

One fine-grid process, then aggregate

1. **Matérn:** Generate once on 48×48 and average to coarser supports. This is the correctly specified baseline.
3. **CAR/ICAR:** Generate once on 48×48 and average. Fitted parameters describe the Matérn approximation to an aggregated GMRF.

Define a new model on each grid

2. **Matérn:** Generate directly at observed-grid locations. The fitted areal model intentionally uses a different support covariance.
4. **CAR/ICAR:** Define a new graph model at each resolution. Both the graph and its implied physical dependence change.

Observed grids have dimensions $2, 3, 4, 6, 8, 12, 16, 24$, and 48 per side. Matérn studies use all 27 combinations of $\sigma^2 \in \{1, 4, 9\}$, $\rho \in \{1/10, 1/3, 1\}$, and $\nu \in \{1/2, 3/2, 5/2\}$. CAR studies use dependence $\alpha \in \{0.75, 0.95\}$ and a separately constrained ICAR model. Each setting has 50 replicates.

How Matérn MLEs are currently obtained

- Given observed $\mathbf{y}_A$, construct $V_\theta = K\Sigma_\theta K^\mathsf{T}$ for candidate Matérn parameters.
- Under $\mathbf{y}_A \sim N(\mathbf{0}, V_\theta)$, minimize the negative Gaussian log-likelihood (up to constants):

$$\log|V_\theta| + \mathbf{y}_A^\mathsf{T} V_\theta^{-1}\mathbf{y}_A. \tag{4}$$

- We currently use R's `optim()` function to minimize this objective jointly over $\theta = (\sigma^2, \rho, \nu)$.
- **Planned improvement:** evaluate the likelihood with a Cholesky factorization and, because $V_\theta = \sigma^2 R_{\rho,\nu}$, profile out σ^2 before optimizing over (ρ, ν).

4. Matérn Simulation Results

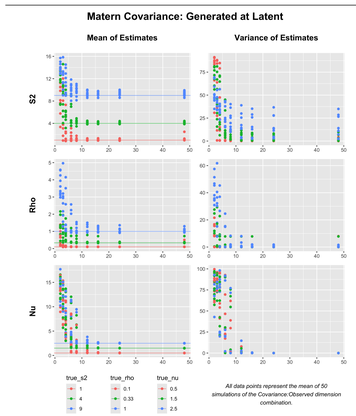


Figure 3. Simulation 1: one latent Matérn process, aggregated to multiple resolutions.

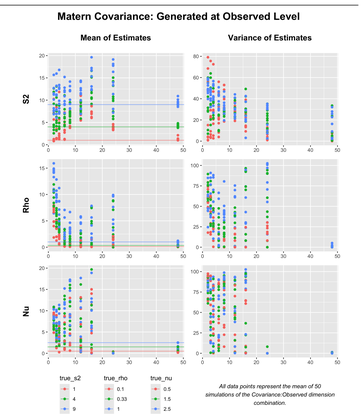


Figure 4. Simulation 2: a Matérn model generated directly at each observed resolution.

Initial findings:

- In Simulation 1, estimates are approximately centered on the generating values; uncertainty increases sharply on the coarsest grids.
- In Simulation 2, the generating point-level covariance and fitted areal covariance differ. Estimates do not consistently recover the generating parameters and are more variable at many resolutions.
- The contrast between Simulations 1 and 2 is consistent with an effect of fitting the same $K\Sigma_\theta K^\mathsf{T}$ model to data generated from different observed-level covariances.

5. CAR/ICAR viewed through an areal Matérn model

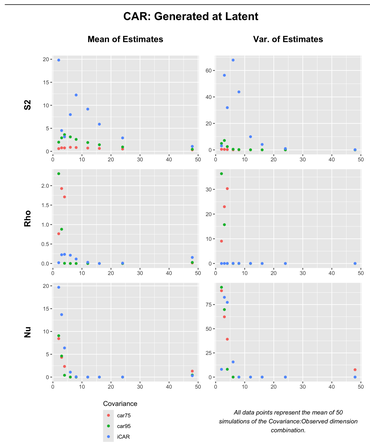


Figure 5. Simulation 3: one latent CAR/ICAR model, then aggregate.

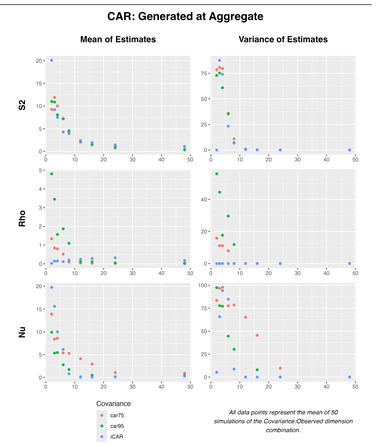


Figure 6. Simulation 4: a new CAR/ICAR model defined on each grid.

Preliminary findings:

- The estimated Matérn parameters summarize CAR/ICAR-generated data through the assumed $K\Sigma_\theta K^\mathsf{T}$ covariance model.
- Across the displayed settings, fitted $\hat{\rho}$ and $\hat{\nu}$ appear to move toward 0 on finer observed grids. Coarse-grid estimates are highly variable.

Connections to existing work:

- Lindgren et al. (2011) establish explicit links between Matérn Gaussian fields and particular GMRF constructions.
- Under specific parameterizations, scaling, and limiting conditions, increasingly fine ICAR models have been related to a Matérn limit with $\rho \to \infty$ and $\nu \to 0$ (Paciorek, 2013).
- Our initial small-$\hat{\rho}$, small-$\hat{\nu}$ pattern motivates further investigation; it should not yet be interpreted as a contradiction of those limiting results.

6. Current conclusions

- Simulation 1 approximately recovers the generating parameters and provides a correctly specified baseline for the areal Matérn likelihood.
- In Simulations 2–4, the fitted Matérn parameters summarize the observed covariance through the assumed $K\Sigma_\theta K^\mathsf{T}$ family; they should not be interpreted as recovering the parameters of their generating models.
- The CAR/ICAR patterns are preliminary and require additional investigation before drawing conclusions about limiting behavior.

7. Next steps

- **Improve and validate estimation:** implement Cholesky-based likelihood evaluation and profile out σ^2; compare with the current `optim()` results and inspect convergence and likelihood surfaces.
- **Compare covariance structures:** at each resolution, compare the true covariance of the observed data with the fitted $K\Sigma_\theta K^\mathsf{T}$ covariance, rather than relying only on parameter estimates.
- **Directly test the centroid approximation:** generate as in Simulation 1, then compare the correct areal likelihood with a point-referenced Matérn likelihood evaluated at region centroids as the support becomes coarser.
- **Investigate CAR/ICAR resolution effects:** determine whether the preliminary patterns in $\hat{\rho}$ and $\hat{\nu}$ persist after checking grid spacing, boundary treatment, and ICAR constraints; also compare the implied correlation structures.
- **Extend and apply:** add intermediate grids, then study resolution sensitivity in ERA5 and other Earth-science data products.

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